



WARWICK BUSINESS SCHOOL
THE UNIVERSITY OF WARWICK

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Question Attempted	Choose one of the AI-based HR tools that we have analysed during the module. Using Tambe, Cappelli and Yakubovich's (2019) AI life-cycle model as a framework, critically examine the potential problems, risks and unintended consequences that may arise at each stage of the AI life cycle when this tool is implemented in HR practice. Use additional literature from the reading list and quality literature that you identify yourself to sharpen your answer. Finally, discuss how—and under what organisational conditions—you would personally use this tool at all, and if so, how you would mitigate the identified risks.
Have you used Artificial Intelligence (AI) in any part of this assignment?	Yes
If you have ticked “Yes” above, please briefly outline below which AI tool you have used, and what you have used it for. Please note, you must also reference the use of generative AI correctly within your assessment, in line with the guidance provided in your student handbook. I have used generative AI tools to support the refinement of this assignment. Specifically, AI was used to assist with improving clarity of expression, polishing wording, and checking grammatical accuracy. It also helped in articulating and structuring my own ideas more clearly. The core ideas, analysis, and arguments presented in this assignment are entirely my own and were developed independently based on my understanding of the module content and the interview conducted. No AI-generated content has been used without critical evaluation and adaptation. AI tool used: Claude	

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Algorithmic Hiring and the Illusion of Objectivity:**A Critical Examination of AI Video Interview Analysis Through Tambe, Cappelli and Yakubovich's (2019) AI Lifecycle Framework****Introduction**

Artificial intelligence (AI) video interview analysis (AVIA) tools, exemplified by platforms such as HireVue, represent one of the most consequential, and contested, applications of machine learning in contemporary human resource management. By automatically scoring candidates based on facial micro-expressions, vocal intonation, linguistic patterns, and eye-contact frequency, AVIA promises to render large-scale candidate assessment faster, cheaper, and purportedly more consistent than human interviewers (Hickman et al., 2022). For large multinational companies, where there may be tens of thousands of applications to sift through across various locations, the advantages are self-evident.

However, such an appeal is based on several premises concerning the legitimacy of psychometric proxies, the neutrality of training data sets, and the ability of human supervision to offset algorithmic bias, none of which stands up to scrutiny. This essay argues that AVIA, rather than eliminating the subjectivity of human judgement, encodes, obscures, and institutionally legitimizes it through the authority of algorithmic outputs. Drawing on Tambe, Cappelli and Yakubovich's (2019) AI lifecycle framework, it critically examines the specific risks, unintended consequences, and mechanisms of harm that emerge at each stage of AVIA's development and deployment. The essay concludes with an argument for the conditional use of AVIA in multinationals, subject to certain mitigating measures, while acknowledging the existence of irreducible risks when using an inherently invalid measure.

Stage 1:**Data Generation: Encoding the Past as the Future**

Tambe et al. (2019) identify data generation as the foundational stage of the AI lifecycle, recognizing that the quality, representativeness, and labelling logic of training data are determinative of everything that follows. In case of AVIA tools, this stage poses even more danger since datasets used to train predictive algorithms are usually based on internal records of recruitment within the company: who was selected, who performed well, and who was promoted. The critical problem is that these records are not neutral historical facts, they are artefacts of the managerial preferences, cultural norms, and structural inequalities that characterized hiring decisions at the time they were made (O'Neil, 2016).

The Amazon recruitment AI case is instructive. According to Dastin (2018), the algorithm used for screening the resumes of potential employees, which had been trained on a decade of past hiring practices, disproportionately discriminated against applicants using the term 'women's' and penalized graduates of all-female universities. It was not because of the racial discrimination of the programmers behind the model but rather that the machine learning algorithm replicated patterns based on its experience, in this instance, learning that successful Amazon engineers were usually men. The same phenomenon applies in AVIA. In the case where the best-performing applicants were predominantly white, male, and outgoing, as decided by the manager, the algorithm will learn to favor such traits regardless of whether they contribute to job success (Barocas and Hardt, 2016).

This is not merely a problem of insufficient data, as is sometimes argued by AVIA vendors who advocate for larger and more diverse training datasets (HireVue, 2021). Barocas and Moritz (2016) have theoretically proven that more data, generated by the same biased procedures, will only exacerbate the existing discrimination. The label of 'successful hire' is given to someone who is favored by a certain manager in certain conditions, not a consistent standard for evaluating job-related skills. When it comes to huge corporations with multiple subsidiaries scattered around different geographical locations, the issue becomes even more complex since the concept of 'high performance' could be culturally relative. For example, while assertiveness and confidence may be positively perceived in North America, the same features could be considered as rudeness or aggression in East Asia or the Middle East. Yet, in a globally trained AVIA algorithm, all these signs would have the same weight (Liem et al., 2018). Thus, the corporation is not only reinforcing the

biases of its subsidiaries but also creating a single global standard, which systematically discriminates against candidates displaying culturally acceptable behavior.

A further, underappreciated risk at this stage concerns what might be called the 'ghost labeller' problem. Where organisations lack the longitudinal performance data to construct reliable outcome labels, AVIA vendors may use proxy measures, such as self-reported job satisfaction, voluntary turnover, or peer ratings, as stand-ins for 'success'. These proxies are themselves contaminated by structural factors (workers in precarious conditions may report lower satisfaction regardless of competence), and their use as ground-truth labels introduces a layer of construct invalidity that is invisible to the HR practitioner who is presented only with the model's output score (Tambe et al., 2019).

Stage 2:**Prediction and Model Building: Contested Science and the Black Box**

The second stage of the lifecycle by Tambe et al. (2019), modeling and predicting, seems to pose the biggest question to the legitimacy of AVIA: the scientific validity of the fundamental psychometric assertions made by AVIA. Typically, AVIA programs use one form or another of Facial Action Unit (FAU) Coding, based on the psychological theory proposed by Paul Ekman that people display their emotions by means of universal facial expressions regardless of culture (Ekman, 1972). The upshot is that a computer program can evaluate a candidate's emotions by looking at how specific facial muscles work in an interview setting.

That bedrock is deeply unstable. Barrett et al. (2019), in a comprehensive review of over a century of psychological research on facial expression, conclude that facial movements do not reliably signal emotional states across individuals, contexts, or cultures. The connection between a smile and happiness, or a furrowed brow and concentration, is neither universal nor deterministic; it is shaped by social context, individual variation, and cultural convention. The use of facial action sequences as predictors of employment-related attributes thus relies on an empirically unjustifiable theoretical underpinning. Hickman et al. (2022) provide a less dramatic assessment, citing empirical data that some of the AVIA scores have some degree of criterion validity as compared to structured interviews. This, too, should be qualified; the significant correlation between some AVIA scores and human interview scores could be due to common method variance between the two measures (both capture impression management). Furthermore, it would be valuable to conduct longitudinal studies investigating whether AVIA scores can predict job performance years into the future after hiring.

The second major risk at this stage is what Wachter, Mittelstadt and Russell (2017) term the 'explainability deficit': the systematic inability of end users to understand, interrogate, or contest algorithmic outputs. Most enterprise AVIA platforms operate as proprietary black boxes. HR practitioners receive a composite score and, occasionally, a high-level explanation (e.g., 'strong communication skills'), but are given no access to the underlying model architecture, feature weights, or decision logic. This is not incidental to the commercial model, it is constitutive of it. Vendors have strong financial incentives to protect their intellectual property. The consequence for large multinationals is that they are purchasing an instrument whose validity they cannot independently verify and whose

outputs they cannot fully explain to a rejected candidate, a trade union representative, or an equality regulator (Raghavan et al., 2020).

Stage 3:**Decision-Making: Automation Bias and the Collapse of Human Judgement**

Tambe et al. (2019) identify the decision-making stage as the point at which algorithmic outputs are translated into HR actions, and it is here that the theoretical promise of 'human-in-the-loop' oversight most conspicuously fails to materialise in practice. The dominant assumption embedded in the deployment rationale for AVIA is that human reviewers will treat algorithmic scores as one input among many, retaining the agency to override, contextualise, and ultimately make the final hiring decision based on a holistic assessment. The empirical literature on automation bias tells a markedly different story.

Goddard, Roudsari and Wyatt (2012) define automation bias as the tendency for human decision-makers to over-rely on automated system recommendations, even when those recommendations conflict with their own assessment or with available contradictory evidence. In the context of AVIA, this dynamic is structurally incentivised: a recruiter who overrides a low AVIA score to progress a candidate who subsequently underperforms bears personal reputational risk ('I ignored the algorithm'), while a recruiter who defers to the algorithm's recommendation is insulated from accountability by the appearance of objective process. The institutional incentive, in other words, runs towards deferral rather than deliberation. Binns (2018) frames this as a problem of 'laundered accountability': the algorithm provides a decision-laundering mechanism that transfers moral responsibility from individual actors to an opaque technical system, thereby reducing the psychological pressure on any specific person to exercise critical judgement.

The power asymmetry introduced by AVIA at this stage deserves particular attention as an underappreciated risk. Candidates subjected to AVIA have no visibility into what is being assessed, how their performance is being scored, or why they have been rejected. This is not merely an ethical concern; it constitutes a structural power asymmetry that fundamentally alters the candidate-employer relationship. In conventional interview processes, candidates can seek feedback, identify areas for development, and build an informed picture of organisational culture through the interview interaction itself. AVIA forecloses all these epistemic possibilities. The candidate is assessed by a system they cannot interrogate, on criteria they cannot access, against benchmarks derived from a population they have no knowledge of. The EU AI Act's (2024) requirement for transparency and the right to human review of high-risk AI decisions partially addresses

this asymmetry, but only in jurisdictions where it applies and only where organisations implement it in good faith rather than as a compliance formality.

A counterargument frequently advanced by AVIA proponents is that algorithmic assessment reduces the in-person biases, affinity bias, halo effects, and interviewer self-similarity preferences, that characterize conventional interview processes (Campion, Palmer and Campion, 1997). This argument is not without merit: unstructured human interviews are themselves deeply unreliable predictors of job performance, with a meta-analytic validity of approximately $r = 0.38$ under favorable conditions (Schmidt and Hunter, 1998). The critical question is not whether AVIA is better than the worst available alternative, but whether it is better than the best available alternative, namely, well-designed structured interviews with trained interviewers and standardised scoring rubrics. The answer to this latter comparison remains, on the available evidence, uncertain at best and negative at worst.

Stage 4:**Implementation and Feedback: Loops, Deskilling, and Regulatory Exposure**

The final stage of Tambe et al.'s (2019) lifecycle, implementation and the subsequent feedback of outcomes into model retraining, generates risks that are both systemic and cumulative. Three interconnected dynamics deserve sustained critical attention.

First, the feedback loop risk. In most instances, when models are deployed in an enterprise setting, their accuracy is regularly checked, and the models are updated using the latest data available. If the primary signal used for retraining is whether AVIA-selected candidates subsequently performed well according to internal performance management systems, the model is iteratively reinforced by the very outcomes it has itself shaped. Candidates who were rejected based on AVIA scores never enter the organisation, and therefore never generate the counterexamples that might reveal the model's errors. This is the closed-loop problem identified across algorithmic systems by Eubanks (2018): the system's predictions become self-fulfilling, because the feedback data available for evaluation is structurally limited to the cases the system approved. Over time, these dynamic narrows the cognitive and demographic range of hired populations in ways that are statistically invisible to internal HR analytics.

Second, adverse selection and gaming effects represent a subtler but equally significant unintended consequence. As more organizations implement AVIA technologies and the public becomes more aware of the technology, especially among the graduating class of major multinational corporations, candidates will game the system to be selected based on the performance criteria set by the AVIA software. In fact, coaching services for candidates on how to adjust their gaze, speech speed, and facial expressions have already emerged in the market (Raghavan et al., 2020). The result is adverse selection; the software selects not candidates with the traits it is supposed to screen but candidates who can play the game of the algorithm.

Third, and perhaps the most consequential aspect of all, is the potential impact on skill levels within the organization. As AVIA embeds itself into the standard procedures of the organization's recruitment practices, the ability to interview independently and well will decline due to lack of use. This is consistent with broader research on technology-induced skill degradation in knowledge work (Autor, Levy and Murnane, 2003). The dependence that the organization develops upon using such technologies becomes not only operational, but strategic, as a multinational which neglects to maintain its ability to

perform interviews becomes locked into the technology in question. The multinational can no longer easily return to a system of independent assessment should there be a problem with the vendor, or a change in laws or public opinion concerning the use of AVIA.

Personal Position: Conditional Use, Governance Architecture, and the Limits of Mitigation

Having examined the risks across all four lifecycle stages, the question of whether to deploy AVIA in a large multinational context admits only a conditional answer. An unconditional refusal would fail to engage seriously with the genuine limitations of the status quo: unstructured human interviewing is neither fair nor particularly valid, and the scale demands of large multinational recruitment cycles create real pressure for process innovation. An unconditional endorsement would, as the foregoing analysis demonstrates, be intellectually indefensible. The responsible position is a narrowly bounded deployment under conditions that are demanding precisely because the risks are severe.

The organisational conditions under which I would consider AVIA deployment are as follows. First, AVIA must function as a structured supplement to, and never a replacement for, human judgement. Its outputs should be treated as one signal in a multi-method assessment process, alongside structured interviews, work-sample tests, and, where appropriate, cognitive ability assessments, instruments whose psychometric properties are substantially better established (Schmidt and Hunter, 1998). Second, the organisation must possess, or commission, genuine independent validation of the vendor's predictive validity claims within its own candidate population, disaggregated by demographic group. Self-reported vendor validity studies, frequently conducted on non-representative samples under favourable conditions, are insufficient for this purpose (Raghavan et al., 2020). Third, the organisation must implement a genuine right of human review and candidate explanation, not merely as a regulatory compliance mechanism under the EU AI Act (2024), but as a substantive commitment to procedural justice.

The governance architecture I would propose encompasses four interconnected mechanisms. An algorithmic audit committee, comprising HR professionals, data scientists, legal counsel, and independent employee representatives, should review model outputs and demographic impact data on a quarterly basis, with mandatory reporting to board level. A pre-deployment bias impact assessment, which is like human rights impact assessments currently being performed by organizations in their sustainability reports, needs to be carried out prior to the deployment and retraining of any new AVIA model. The vendor agreements must involve strict data transparency requirements with regards to model documentation, which would allow carrying out the independent audit, and it is an absolute must from the point of procurement perspective.

It is necessary, however, to be honest about the limits of these mitigation strategies. Governance frameworks are effective only insofar as they are genuinely operationalised, adequately resourced, and insulated from the commercial pressures that typically drive AVIA adoption in the first place. Organisations that adopt AVIA primarily to reduce time-to-hire and cost-per-hire are structurally unlikely to invest the resources required for rigorous ongoing auditing. More fundamentally, no governance structure can resolve the foundational problem identified at Stage 2: if the scientific basis for inferring personality traits from facial and vocal cues is itself contested, as Barrett et al. (2019) compellingly argue, then governance can make the deployment more procedurally fair but cannot make it more predictively valid. The instrument may remain fundamentally unsuited to its stated purpose regardless of how responsibly it is governed. The Chartered Institute of Personnel and Development's (CIPD, 2020) call for algorithmic transparency in recruitment is a step in this direction but falls short of the structural interventions that the severity of the risks described here would justify.

Conclusion

AVIA embodies a paradox that lies at the heart of AI's introduction into HRM. The use of AI makes possible the removal of bias from the evaluation process through the adoption of an objective approach but sacrifices accountability as a trade-off. Through the conversion of a subjective evaluation process to a numeric scoring algorithm, the human factor behind this type of discrimination becomes invisible in the former but magnified in the latter as the unseen training data set, algorithm structure, and the feedback loop. An analysis using Tambe et al.'s (2019) lifecycle can reveal the specific instances in which such risks become prevalent. These can occur in the form of the inclusion of past discrimination through biased data creation, contentious psychological concepts being re-coded into proxies, and automation bias reducing human decision-making ability.

In a large multinational firm, however, these factors do not merely compound; they multiply. The same scale that makes AVIA appealing financially magnifies the demographic discrepancies, intensifies the gaming problem, and heightens the dependency on a vendor who cannot prove his own veracity. An application under such constraints can be justified when the psychological measurement tool is valid, when the sample is representative, when the oversight is real, and when the organization's adherence to the principle of justice is structural and not just rhetorical. Under current technological circumstances and institutional context, these four conditions tend not to coexist. The responsibility for proving an affirmative answer remains with the party deploying the system, and, from our review of the evidence above, that responsibility has yet to be met.

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